Customer Churn Prediction

1. Overview

This case analyzes churn trends among 20,000 customers using a logistic regression model. From the total customer base, 34.2% (6,843 customers) were recorded as having churned, while 65.8% (13,157 customers) remained. The analysis focused on the relationship between the number of calls to customer service (*support_calls*) and the likelihood of customers discontinuing their subscription.

2. Dataset Description

The dataset contains information on 20,000 customers of a subscription-based service company (telecommunications), with the following variables:

Variable	Description
customer_id	Unique identifier for each customer
tenure	Number of months the customer has been subscribed to the service
monthly_charges	Amount billed to the customer each month
total_charges	Total amount billed to the customer over their tenure
contract	Contract type (Month-to-month, One year, Two year)
payment_method	How the customer pays their bill (Credit, Debit, Cash, UPI)
internet_service	Type of internet service subscribed (DSL, Fiber, or none)
tech_support	Whether the customer subscribes to tech support (No/Yes)
online_security	Whether the customer subscribes to online security (No/Yes)
support_calls	Number of times the customer contacted customer support
churn	A factor with levels No and Yes indicating whether the customer churned

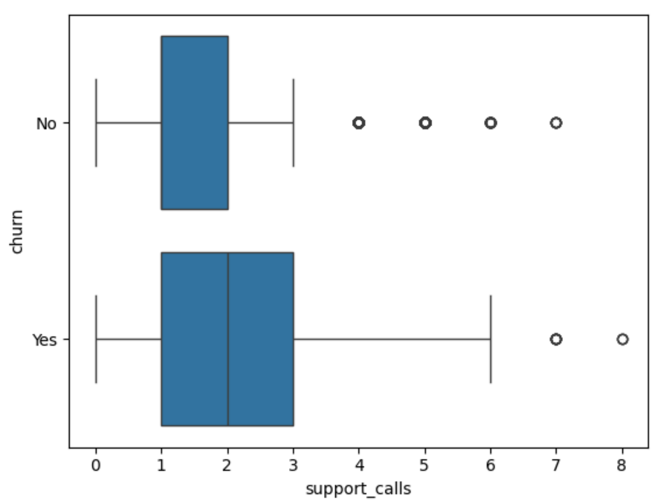
3. **Exploratory Data**

- Comparison of Group Means

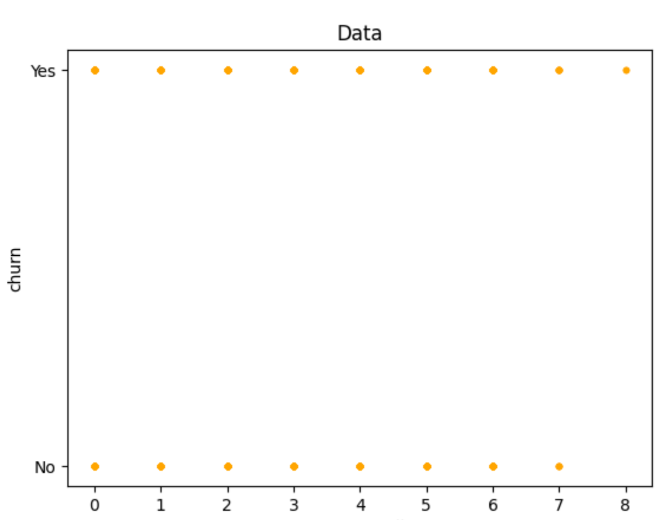
Metrics	No Churn	Churn
Average support calls	1.34	1.85
Average monthly_charges	64,912	79,814
Average tenure	37,8 months	33,9 months
Average total_charges	2,454,145	2,716,702

Customers who churn consistently have a higher average number of support calls and larger monthly bills, yet a shorter tenure compared to customers who remain. This pattern reinforces the hypothesis that the frequency of support calls is linked to customer dissatisfaction that leads to churn.

- Distribution of Support Calls by Churn Status. The boxplot shows that customers who churn tend to have a higher number of support calls, with greater dispersion, compared to customers who do not churn.



- Raw Data Distribution. The scatter plot reveals a natural non-linear pattern between the number of support calls and churn status (yes/no)



4. **Data Preparation and Methodology**

The churn target variable, originally categorical (Yes/No), was converted to a binary numeric format using Label Encoding (No = 0, Yes = 1). No missing values were found in the *support_calls* or *churn* variables; thus, the data was ready for modeling without further processing.

5. **Model Results**

The model used is logistic regression, which models the probability of churn as a logistic function of a linear combination of predictor variables:

$$P(\text{churn}) = \text{logit}^{-1}(\beta_0 + \beta_1 \times \text{support\_calls})$$

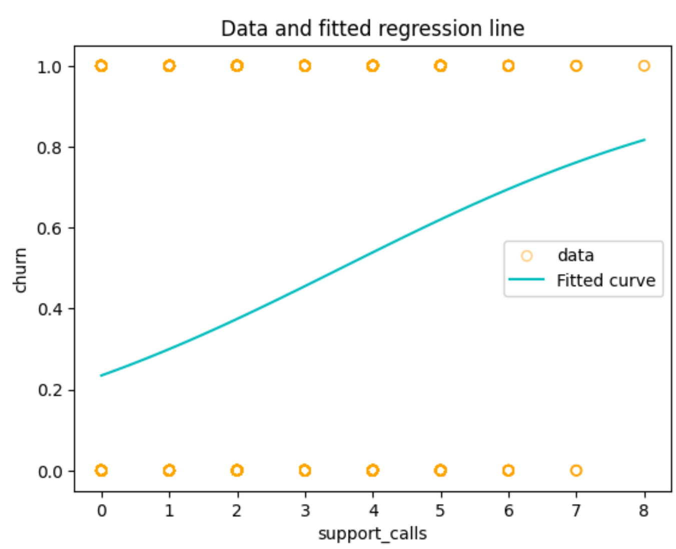
Coefficient estimates were obtained using the maximum likelihood estimation (MLE) method. The model estimation results are as follows:

Parameter	Coefficient	Standard Error
Intercept	-1.1823	0.0252
support_calls	0.3345	0.0123

So the final model formed is:

$$P(\text{churn}) = \text{logit}^{-1}(-1.182 + 0.335 \times \text{support\_calls})$$

Additional model statistics: both coefficient values are statistically significant (p-value < 0.001), with a pseudo R^2 of 0.030 and a log-likelihood of -12,466.16. The odds ratio for *support_calls* is $e^{0.335} \approx 1.40$. Churn probability curve derived from the logistic regression model against actual data.



6. **Interpretation of Coefficients**

- The positive coefficient for *support_calls* indicates that the more frequently a customer contacts customer support, the higher the probability of that customer churning.
- Specifically, each additional customer support call is associated with a 0.335-unit increase in the log-odds of churn.

- Using the "divide-by-4" rule: near the average number of calls, each additional customer support call is associated with an increase in churn probability of approximately 8.4%.
- In terms of the odds ratio, each additional customer support call increases the odds of churn by a factor of approximately 1.40 (40% higher) compared to the odds prior to that additional call.

7. **Example Prediction**

To illustrate, the model is used to predict the churn probability for two customers with a difference of four customer support calls (one call versus five calls):

Number of Support Calls	Churn Probability
1 times	0,2999 (~30,0%)
5 times	0,6202 (~62,0%)

Customers who have contacted customer support only once have a churn probability of approximately 30%. In contrast, customers who have contacted customer support five times have a significantly higher churn probability—around 62.0%.

8. **Conclusions and Recommendations**

- The number of calls to customer support has proven to be a statistically significant predictor of the likelihood of customer churn.
- Customers with a high frequency of customer support calls face a significantly higher risk of churn compared to those with a low call frequency.
- It is recommended that the company establish an early warning system to monitor customers who exceed a specific threshold of customer support calls (e.g., more than 3–4 calls), enabling the retention team to intervene at an earlier stage.
- As this model relies on a single predictor variable (*support_calls*) and yields a relatively low pseudo-R^2 (0.030), it is advisable to develop an advanced model incorporating additional variables—such as tenure, *monthly_charges*, *contract*, *tech_support*, and *online_security* —to improve prediction accuracy.